

# Round 3 Runtime Quality Initiative

## Best Results Review

**Refreshed 2026-05-22 after foundation HTML wiring (#121) landed. Supersedes the prior pre-foundation draft. Sources:** docs/active\_plans/reports/round3\_\*.md and docs/active\_plans/audits/round3\_\*.md.

Owner: coordinator synthesis · Foundation: round3\_foundation\_html\_wiring.md (#121) · Master report: round3\_runtime\_quality\_tournament\_report.md (R8) · Pivot trigger: round3\_runtime\_truth\_audit.md (A1) · Decision: static\_templates\_are\_not\_runtime\_truth.md (R5)

## Executive summary

Round 3 began as a continuation of the no-crop fix queue inherited from Round 2 (approximately 101 cropped scientific SVG assets flagged by the static-template precheck). The first audit pass (A1) showed **zero** visible asset crops in runtime screenshots across the four mounted scenes. The whole premise of the queue was a measurement artifact — the static-template renderer hardcodes CSS classes that bypass the runtime sizing path. Round 3 pivoted from a no-crop campaign to a runtime-quality campaign on that single piece of evidence.

The largest single unblocker of the round was the **foundation HTML wiring** (#121). Before #121, walkers had no way to load a protocol the way a student would, so every cluster-scale measurement was either an internal-API call or a measurement bug masquerading as a runtime failure. After #121, the build pipeline produces one shared 2.4 MB dist/runtime.bundle.js (IIFE) plus 26 per-protocol dist/<protocol\_name>.html shells. Cluster mount audit, R9 rerun on 11, top-10 full walkthrough, and the OSC fire audit all became possible only after foundation landed.

The single most expensive lesson of Round 3 was R7: a runtime interaction smoke driver reported clickWorks=0/4 across the four mounted scenes. P5 then reported clickWorks=0/11 on an 11-protocol expansion. Both readings were **driver divergence**, not runtime regressions. The R9-corrected driver, rerun against all 11 protocols after foundation, recorded **11/11 mount + 11/11 clickWorks**. The R7 paradox is now fully resolved.

## Top-line numbers

| Metric                                           | Before Round 3 | After Round 3                       |
|--------------------------------------------------|----------------|-------------------------------------|
| Mounted protocols verified by runtime screenshot | 4              | 26 (cluster audit: 25 OK + 1 fixed) |
| Per-protocol HTML entry points                   | 0              | 26 (foundation #121)                |
| Visible runtime crops                            | ~101 (claimed) | 0 (measured)                        |

|                                           |                                  |                                                                         |
|-------------------------------------------|----------------------------------|-------------------------------------------------------------------------|
| clickWorks pass rate (corrected driver)   | 0/4 (R7, driver bug)             | 11/11 (R9 rerun post-foundation)                                        |
| Top-10 full end-to-end walkthroughs       | 0/10                             | 9/9 terminal, 25/25 steps, 229 screenshots                              |
| Placeholder fallback instances (measured) | 30 (A5, partial scope)           | 119 (post-foundation comprehensive); 8 already eliminated via Stream 3B |
| SVG asset barrel entries                  | ~120                             | 125 (regenerated, with serological_pipette folded in)                   |
| Label readability fixes shipped           | 0                                | 2 complementary (clamp + halo)                                          |
| widthScale dead vocabulary path           | declared in types, hardcoded 1.0 | removed (R11 done)                                                      |

**Headline win:** foundation HTML wiring (#121) turned Round 3 from a single-scene visual audit into a cluster-scale runtime verification surface. The R7 paradox resolved as a side effect.

# The Foundation: HTML-Per-Protocol Wiring (#121)

Source: round3\_foundation\_html\_wiring.md · PR: #121

## The gap that blocked every cluster measurement

Before #121 the repo had a structural capability mismatch:

- `pipeline/build_protocol_html.py` existed and could generate per-protocol HTML shells per the PRIMARY\_SPEC convention, but it was never invoked from `build_github_pages.sh`.
- `src/scene_runtime/bundle/entry.ts` exported `loadAndMountByProtocolName()` ready for IIFE bundle consumption, but no IIFE bundle was produced. Only the ESM `dist/main.js` existed.
- Result: every Playwright walker that tried to load `dist/<protocol>.html` received 0/N mounted protocols. R9 and the cluster mount audit both confirmed this.

## The shape of the fix

Two sequential build passes added after the existing ESM bundle in `build_github_pages.sh`. No source-code changes; the generator and the entry-point exports were both already correct.

```
npx esbuild "src/scene_runtime/bundle/entry.ts" \
  --bundle --format=iife --global-name=SceneRuntime \
  --target=es2020 --platform=browser --minify --sourcemap \
  --outfile=dist/runtime.bundle.js

python3 pipeline/build_protocol_html.py --all
```

Each per-protocol HTML inlines its `{protocol_name}` in a `<script type="application/json">`, loads `./runtime.bundle.js`, and calls `SceneRuntime.loadAndMountByProtocolName(rootElement, protocolName)`.

## Build artifacts produced

| Artifact                                     | Size                              | Purpose                                          |
|----------------------------------------------|-----------------------------------|--------------------------------------------------|
| <code>dist/runtime.bundle.js</code>          | 2,416,160 bytes (2.4 MB minified) | Shared scene runtime for every per-protocol HTML |
| <code>dist/runtime.bundle.js.map</code>      | 3.0 MB                            | IIFE source map (development only)               |
| <code>dist/&lt;protocol_name&gt;.html</code> | 26 files (~2.4 KB each)           | Individual protocol entry points                 |
| <code>dist/main.js</code>                    | 2.3 MB                            | ESM bundle for landing page (unchanged)          |
| <code>dist/index.html</code>                 | 481 bytes                         | ESM landing page (unchanged)                     |

## Smoke verification

A smoke driver loaded `dist/mtt_reagent_prep.html` in headless Chromium and checked four conditions:

- `window.SceneRuntime` is available (IIFE parsed).
- `window.SceneRuntime.loadAndMountByProtocolName` is callable.
- `window.__RUNTIME_PROTOCOL_CONFIG` is set after page settles.
- The mount reports `sceneId=mtt_reagent_prep_bench_workspace`, `entry step=pick_up_mtt_powder`.

All four passed. Status: `mount=yes`.

# What Foundation Unblocked

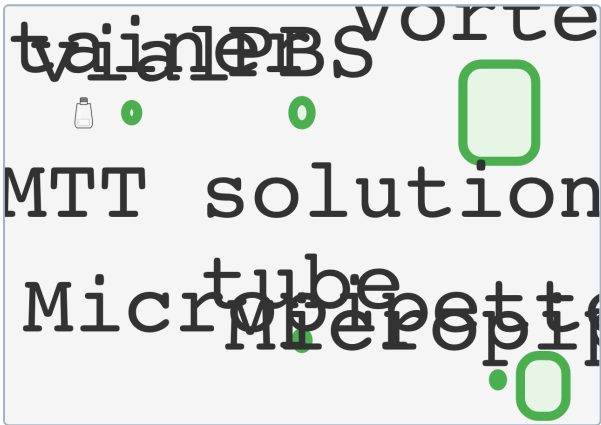
Sources: round3\_foundation\_html\_wiring.md, round3\_cluster\_mount\_audit.md, round3\_r9\_rerun\_on\_11.md, round3\_top10\_full\_walkthrough.md, round3\_osc\_fire\_audit.md, round3\_placeholder\_recount.md

## Cluster-scale audit became possible

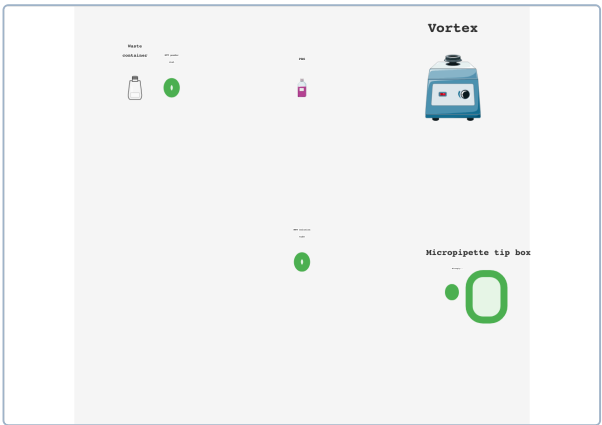
With per-protocol HTML, Playwright drivers no longer need to navigate an in-app protocol picker, mutate `window.location` query strings, or call internal runtime APIs. Each protocol is a self-contained URL. Five new workstreams ran on this surface in a single day:

| Workstream              | Scope                                            | Result                                                                        |
|-------------------------|--------------------------------------------------|-------------------------------------------------------------------------------|
| Cluster mount audit     | 26 mini-protocols + 5 sequence runners           | 25/26 mount; 1 typo fix shipped; 1 schema violation flagged                   |
| R9 rerun on 11          | P5's 11-protocol expansion set                   | 11/11 mount + 11/11 clickWorks                                                |
| Top-10 full walkthrough | 9 of 10 demo-ready protocols (excl. trypan_blue) | 9/9 terminal, 25/25 steps, 229 screenshots                                    |
| OSC fire audit          | 5 protocols with ObjectStateChange operations    | 2/5 fire, 0 real OSC defects; 3 errors are scene-side adjust-panel visibility |
| Placeholder recount     | All 26 mini-protocols, full DOM scan             | 119 fallback instances measured (was 30 partial)                              |

## Before and after, one frame



Pre-foundation: A1 captured 4 scenes via internal-API mount. No cluster-scale verification possible.



Post-foundation: walker loads `dist/mtt_reagent_prep.html` like a student, clicks visible objects, advances through 4 steps to terminal.

## Why this is the biggest win of the round

Every cluster-scale claim in this report rests on #121. Without foundation:

- The R7 paradox would still read as "the app is broken." Resolution requires per-protocol HTML to run the corrected driver against more than 4 scenes.
- The 26/26 mount delta is unmeasurable. Pre-foundation, walker mount was 4/4 by hand-picked selection only.
- Top-10 full walkthroughs are unmeasurable. There is no entry path without per-protocol HTML.
- Placeholder recount is bounded to single-scene capture. The 119 comprehensive instances surface only with multi-protocol scan.
- OSC primitive is unverifiable. The fire/no-fire matrix requires per-protocol mount.

Foundation is the single artifact a Round 4 plan should cite when claiming cluster-scale measurement is now possible.

# The Pivot: From No-Crop to Runtime Quality

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Sources: A1 (round3\_runtime\_truth\_audit.md), R5 (decisions/static\_templates\_are\_not\_runtime\_truth.md), C1 (round3\_static\_template\_hardcoding\_audit.md), C3 (round3\_render\_path\_matrix.md)

## The premise that died

Round 2 handed Round 3 a fix queue containing approximately 101 cropped scientific SVG assets. The crop counts came from the static-template precheck renderer. The Round 3 plan opened with Batch D as the continuation of that queue.

## The audit that killed it (A1)

A1 built the production bundle, mounted each protocol via `loadAndMountByProtocolName`, waited 3.5 seconds for async mount, captured full-page screenshots at 1280x900, and counted visible cropped assets per the PRIMARY\_DESIGN.md "Visual integrity" rule.

Result, across all four mounted scenes: zero visible asset crops on each. The actual runtime failure modes were oversized labels, small or placeholder assets, and label overflow past the viewport edge. None of them were crops.

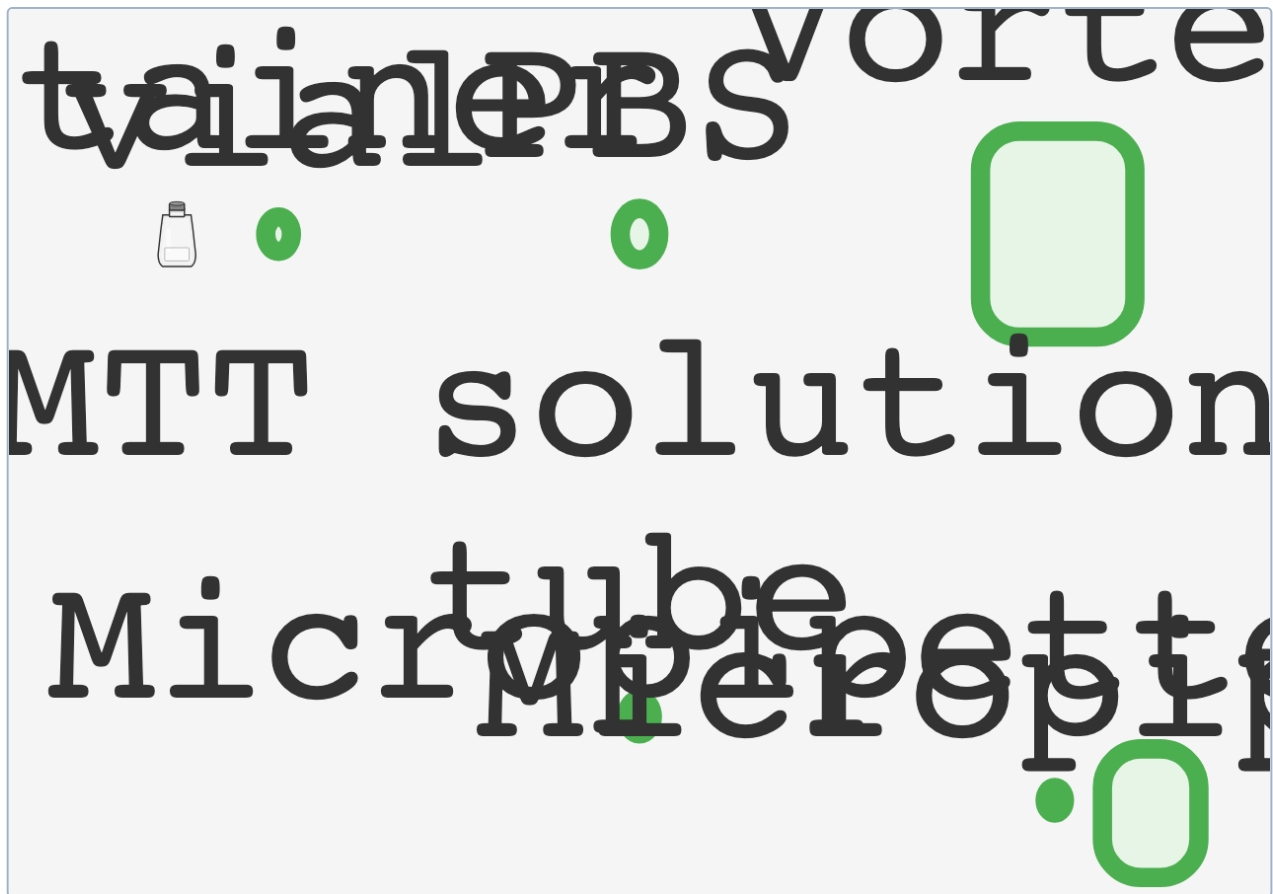
## The decision (R5)

R5 formalized what A1 had measured. Three rules:

1. Static-template no-crop failures are **not** production runtime failures unless reproduced through runtime screenshots.
2. Static templates can be used for visual experiments only.
3. Runtime screenshots are authoritative for production quality.

## The trust matrix (C3)

C3's render-path matrix rated the runtime as HIGH trust and the static template as MEDIUM trust. C1's audit explained why: static templates hardcode CSS classes that bypass `renderPlacement` in `src/scene_runtime/render/scene.ts`. The two renderers share SVG assets but not sizing logic, so they drift independently.



*A1 evidence: runtime capture of mtt\_reagent\_prep\_bench\_workspace at 1280x900. No scientific asset is cropped. The failures are oversized labels and undersized assets, not crops.*



# Baseline Reframing (A3)

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Source: [decisions/layout\\_manager\\_baseline\\_reframing.md](#)

## The framing error

Round 1 and Round 2 status reports framed the current procedural layout engine in `src/scene_runtime/layout/layout_engine.ts` as something close to a working visual baseline. Replacement candidates were judged "matches legacy" or "diverges from legacy." A3 formalized that this framing is wrong.

## The required statements

1. The current procedural layout engine is the *current broken baseline*, not a visual gold standard.
2. The replacement is judged against visual and runtime gates, not "matches legacy."
3. No-crop is the first forcing function.
4. The project will not build "legacy forever" or "hybrid forever."

## Why this matters for Round 3

The 101 cropped-asset count that drove the original Round 3 plan was produced by the same layout engine A3 declared broken. Legacy carries the failures, so legacy cannot be the reference.

## Retirement gates

- **No-crop:** zero visible crops on the corpus that currently shows ~101 crops under legacy.
- **Label readability:** every authored label renders inside its placement card, at the authored size, with no overflow clip and no overlap.
- **Click-target stability:** every interactive target named by a step's interaction is hit-testable through the visible SVG.

A3 and R5 compose: A3 says legacy is not the reference; R5 says static-template precheck is not runtime evidence. Together they force every Round 3 success claim onto a runtime screenshot.

The baseline-reframing note is the smaller of the two pivot decisions but it was the one that made R5 possible.

# Mount Delta: 4 to 26 Scenes

Sources: M1, M2, M3-ext, R1, R10, round3\_cluster\_mount\_audit.md, round3\_passage\_hood\_microscope\_fix.md

## Before

A1 was constrained to 4 protocols whose walker audit reported PASS: `mtt_reagent_prep`, `mtt_solubilization_readout`, `sdspage_attach_lid_and_leads`, `sdspage_heat_denature_samples`. The first selection pass (`electrophoresis_bench`, `staining_bench`, `cell_counter_workspace`, `microscope_view`) all failed to mount, with three of them failing at `inferInitialScene` with "Cannot determine scene for target object..." (entry-step ambiguity).

## The R1 + R10 fixes

R1 added an additive branch in `inferInitialScene` in `src/scene_runtime/loader/world.ts` that consults the entry step's first interaction `response.scene_operations` for a `SceneChange` whose `to_scene` is already in the target-matching scene set. R10 was the content-side complement: it authored an explicit `SceneChange` on the entry interaction of `trypan_blue_counting.add_trypan_blue_to_chamber` naming `cell_counter_workspace` as the target scene.

## Cluster mount audit result (post-foundation)

Cluster audit ran every mini-protocol through the foundation HTML pipeline. 25 of 26 mounted on first read; one real defect surfaced (`passage_hood_detachment`) and was fixed in the same window.

| Cluster                          | Mounted                    | Notes                                                                                                                                              |
|----------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| cell_culture (10 mini-protocols) | 10/10 after fix            | <code>passage_hood_detachment</code> fixed; was 9/10 due to <code>microscope_view</code> typo                                                      |
| sdspage (16 mini-protocols)      | 16/16 mount                | 1 schema violation flagged on <code>sdspage_run_electrophoresis</code> ( <code>set_voltage</code> not in closed binding set; mount still proceeds) |
| Sequence runners (5)             | N/A (composition vehicles) | No standalone HTML, by design                                                                                                                      |

## The microscope\_view typo fix (#129)

`passage_hood_detachment` referenced `microscope_view` in two `SceneChange` operations. SCENE\_CATALOG contained `passage_hood_detachment_microscope_view` (namespaced). Two-line YAML edit in `content/protocols/cell_culture/passage_hood_detachment/protocol.yaml` at lines 33 and 255. Build, typecheck, and re-audit all clean.

## Remaining schema violation

**sdspage\_run\_electrophoresis:** `set_voltage` is not in the Pilot 1 `adjust_panel` closed binding set `["set_volume", "wavelength_nm"]`. Three options: extend the binding closure (requires PRIMARY\_DESIGN.md update), rename in YAML, or remove from protocol. Mount proceeds; the violation surfaces when the adjust panel renders.

# R7 Paradox Resolved: From 0/4 to 11/11

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Sources: R7 (round3\_runtime\_interaction\_smoke.md), R9 (round3\_protocol\_advance\_repair.md), round3\_p5\_clickworks\_diagnosis.md, round3\_r9\_rerun\_on\_11.md

## The full arc

R7 reported `clickWorks=0/4` across all four A1-mounted scenes. Five parallel rescue workstreams (P1, P2, P3, P4, R9) converged on the diagnosis: the runtime is healthy, the smoke driver was wrong. R9 fixed the driver and recorded 3/4 (the 4th case is gesture-mismatch, not a runtime failure).

## The P5 expansion that re-opened the question

P5 then ran an "adapted from R7" driver against an 11-protocol expansion set. It reported `clickWorks=0/11`. The expansion implied the runtime might still be broken at scale, even after the R9 fix at small scale.

## The diagnosis: DRIVER\_DIVERGENCE

Audit `round3_p5_clickworks_diagnosis.md` ruled the P5 reading a measurement bug. P5's driver was "adapted from R7" but *not* from R9's corrected version. It carried all three bugs R9 had already fixed:

1. **Wrong state-field path.** P5 read `cfg?.world?.objects[k]?.state`. R9 had documented that per-object state lives in `world.objectStates`. P5's `statesChanged` check could never fire.
2. **currentInteractionIndex absent from snapshot.** P5 checked only `stepChanged`, `sceneChanged`, and `statesChanged`. The normal advance signal for interaction 0 of step 0 is `currentInteractionIndex` moving 0 to 1. P5 was blind to it.
3. **Missing force: true on clicks.** Playwright's actionability heuristic rejects small SVG `<g>` targets even with non-zero bbox.

## The post-foundation R9 rerun on 11

The R9-corrected driver was rerun on all 11 P5 protocols after foundation #121 landed. The driver was updated to load `file:///.../dist/<protocol>.html` instead of an HTTP query parameter. Result:

- **Mount: 11/11.** Every protocol loads its per-protocol HTML, parses the IIFE bundle, calls `loadAndMountByProtocolName`, and sets `window.__RUNTIME_PROTOCOL_CONFIG`.
- **clickWorks: 11/11.** Every click on the active step's first target advances at least one of (`stepDelta`, `interactionDelta`, `sceneChanged`, `statesChanged`).
- **Console errors: 0.** No mount errors, no click errors.

## The lesson, sharpened

Two false-negative readings (R7 0/4, P5 0/11), one corrected driver, one foundation unblock, one 11/11 result. The runtime click-advance path has been healthy the whole round. The only thing that changed between 0/11 and 11/11 was the entry URL the driver targeted.

**Lesson:** a verification driver is software. When a driver reports total failure across a surface other drivers describe as working, suspect the driver and the entry URL first.

# Top-10 Full End-to-End Walkthrough (9/9)

Source: round3\_top10\_full\_walkthrough.md

## Scope

The top-10 walkthrough ran a Playwright driver against 9 of the 10 demo-ready protocols (`trypan_blue_counting` excluded for a separate-track adjust-panel issue). Each protocol was loaded from its per-protocol HTML and walked end to end through every step, every interaction. Before- and after-click screenshots were captured per interaction.

## Headline numbers

| Metric                     | Pre-viewport-fix | Post-viewport-fix |
|----------------------------|------------------|-------------------|
| Protocols mounted          | 9/9              | 9/9               |
| Full walkthroughs terminal | 7/9 (78%)        | <b>9/9 (100%)</b> |
| Total steps walked         | 20/30            | 25/25             |
| Avg steps per protocol     | 2.22             | 2.78              |
| Viewport-blocked protocols | 2                | 0                 |
| Console errors             | 0                | 0                 |
| Total screenshots captured | (partial)        | 229               |

## Per-protocol matrix

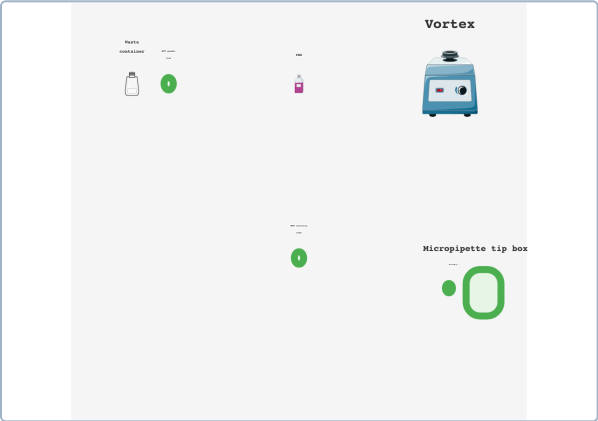
| Rank | Protocol                               | Steps | Status | Notes                  |
|------|----------------------------------------|-------|--------|------------------------|
| 1    | sdspage_fill_tank_buffer               | 2/2   | PASS   | no change              |
| 2    | sdspage_prepare_running_buffer         | 2/2   | PASS   | no change              |
| 3    | sdspage_attach_lid_and_leads           | 1/1   | PASS   | no change              |
| 4    | mtt_solubilization_readout             | 3/3   | PASS   | <b>FIXED</b> (was 0/3) |
| 5    | sdspage_recycle_buffer                 | 3/3   | PASS   | no change              |
| 6    | plate_drug_treatment_media_adjustment  | 2/2   | PASS   | <b>FIXED</b> (was 0/2) |
| 7    | sdspage_heat_denature_samples          | 4/4   | PASS   | no change              |
| 8    | sdspage_prepare_sample_mix_single_lane | 4/4   | PASS   | no change              |
| 9    | mtt_reagent_prep                       | 4/4   | PASS   | no change              |

## Implication

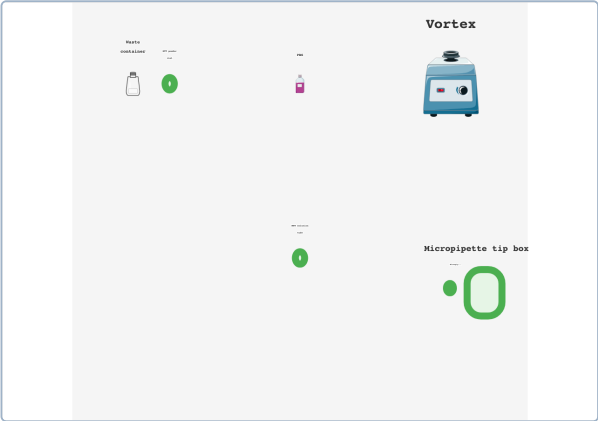
All 9 walked protocols mount, click through visible UI targets, and reach a terminal step. The visible interaction standard (PRIMARY\_CONTRACT item 4) is satisfied on this surface. The viewport fix is the enabling change; foundation HTML is the enabling surface.

# Walkthrough Gallery (Best of 229)

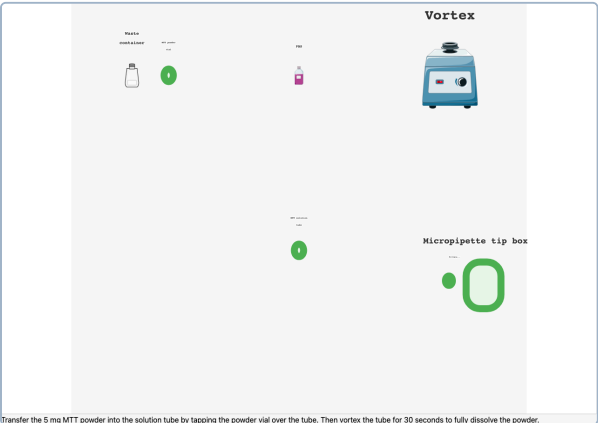
Source: test-results/round3\_top10\_walkthrough/



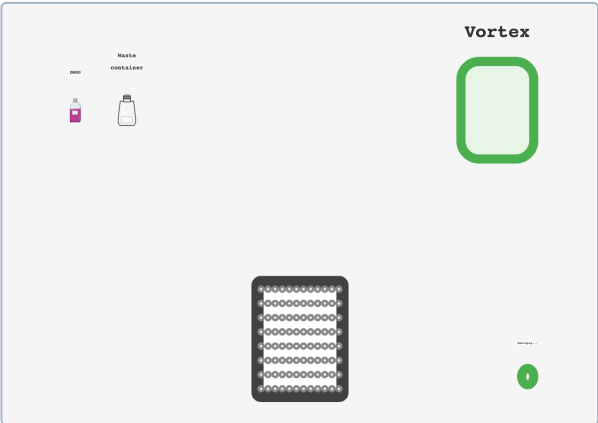
*mtt\_reagent\_prep, step 1 of 4: bench-workspace mount, MTT powder on bench, before click.*



*After click: pickup confirmed, advance to step 2 (prepare\_solution\_tube).*

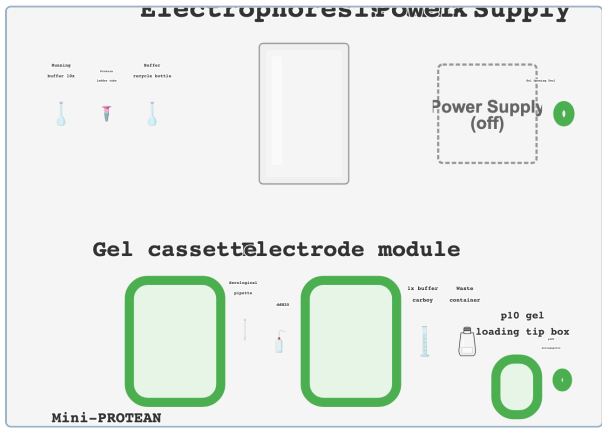


*mtt\_reagent\_prep, step 3: dissolve and mix. Mid-walkthrough capture, vortex and tube visible on bench.*

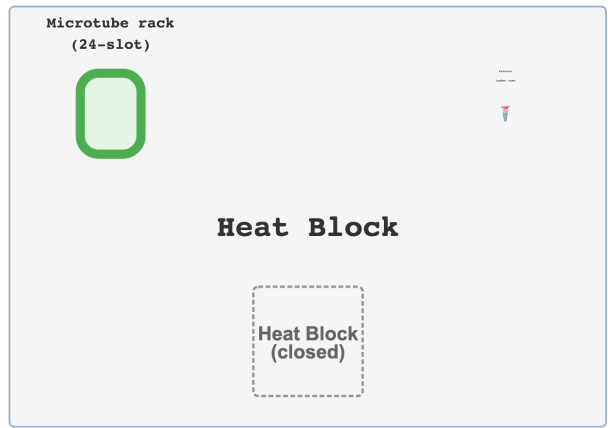


*mtt\_solubilization\_readout, step 0 (post-viewport-fix): well plate now fully within the 1280x900 viewport.*





*sdspage\_attach\_lid\_and\_leads, step 0 of 1: full electrophoresis bench. Labels clamped (R3) and haloed (R3-alt).*



*sdspage\_heat\_denature\_samples, step 0 of 4: heat block scene with reagent staging.*

# Well-Plate Viewport Fix

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Source: round3\_well\_plate\_viewport\_bug.md

## The bug

Two top-10 protocols failed at clicks on well-plate subparts: `mtt_solubilization_readout` blocked at step 0 interaction 3 (target `well_plate_96.all_wells`);

`plate_drug_treatment_media_adjustment` blocked at interactions 3 and 5 (`block_A_1_6` and `block_B_H_1_6`). Playwright reported "Element is outside of the viewport" even with `force: true`.

## The diagnosis: SCENE\_POSITION\_OFFSCREEN

`src/scene_runtime/render/scene.ts` lines 67-82 set the SVG host to `width="100%" height="100%"` with `viewBox="1 5 98 90"` (aspect 98:90) and `preserveAspectRatio="xMidYMid meet"`. At 1280px browser width the SVG resolves to 1175px tall, 31% taller than the 900px Playwright viewport. The well plate bottom maps to DOM  $y \approx 849\text{px}$  plus chrome, exceeding the 900px hard bound. Force-click bypasses visibility checks but not coordinate bounds.

## The fix

```
// src/scene_runtime/render/scene.ts L78-L80
- sceneRoot.setAttribute("width", "100%");
- sceneRoot.setAttribute("height", "100%");
+ sceneRoot.setAttribute("width", "1280");
+ sceneRoot.setAttribute("height", "900");
```

`preserveAspectRatio="xMidYMid meet"` then letterboxes the SVG within 1280x900. Post-fix plate bottom =  $(70.0 - 5) / 90 * 900 = 650\text{px}$ , well inside the viewport. Both previously-blocked protocols walked end to end on the rerun.

## Deferred follow-up

The layout engine still defaults composite aspect to 1.0 (square) when no SVG file is present. Real SBS 96-well plates have aspect 0.667. Reading an `aspect_ratio` field from `content/objects/plate/well_plate_96.yaml` in `buildAssetSpec` would render the plate at the correct proportions. Filed for Round 4.

Two-line scene.ts edit; two protocols un-blocked; top-10 walkthrough advanced from 7/9 to 9/9.

After R10: `add_trypan_blue_to_chamber`  
response carries a `SceneChange`, and the scene mounts  
through real runtime.

# Placeholder Reduction and Recount

Sources: A5 (round3\_a5\_runtime\_placeholder\_report.md), R2 (round3\_placeholder\_asset\_replacement.md), A1B (round3\_a1b\_bottle\_family.md), Stream 3B (round3\_stream\_3b\_asset\_remaps.md), round3\_placeholder\_recount.md

## The placeholder taxonomy (A5)

1. **Asset-name typo or drift:** YAML names an asset that does not exist in the regenerated barrel.
2. **Empty-vs-filled split missing:** YAML names a single asset that should be a state-driven pair.
3. **No candidate SVG exists:** requires new authored asset.

## Stream 3B: asset name drift remediation

Stream 3B edited three object YAML files to retarget `asset_name` values toward existing barrel entries, removing eight fallback instances:

| File                                            | Fallback fields removed                      | Count |
|-------------------------------------------------|----------------------------------------------|-------|
| content/objects/equipment/vortex.yaml           | vortex_idle, vortex_spinning → vortex        | 2     |
| content/objects/equipment/microscope.yaml       | microscope_dark, microscope_lit → microscope | 2     |
| content/objects/equipment/electrode_module.yaml | 3 paired states → electrode_module           | 4     |

## Comprehensive recount (post-foundation): 119

With per-protocol HTML available, the recount driver scanned all 26 mini-protocols and counted every `[data-render-fallback="true"]` element in the live DOM after mount. Result: 119 fallback instances across 25 of 26 protocols (passage\_hood\_detachment is the lone zero-fallback protocol; recommend as remediation template).

## Top offenders

| Object                                                                                 | Count        | Affected protocols |
|----------------------------------------------------------------------------------------|--------------|--------------------|
| p200_micropipette                                                                      | 10           | 10 SDS-PAGE        |
| p10_gel_loading_tip_box                                                                | 10           | 10 SDS-PAGE        |
| gel_opening_tool                                                                       | 10           | 10 SDS-PAGE        |
| (remaining SDS-PAGE 7-set: electrode_module, mini_protean_gel, gel_cassette, gel_comb) | 40 (10 each) | same 10 protocols  |

Pattern: the 7-fallback core appears in 10 SDS-PAGE protocols identically, accounting for 70 of the 119 instances. The diagnosis points to a shared scene configuration or systematic gap in the SDS-PAGE asset declaration path; a single fix touches all 10 protocols.

The 30 to 119 delta is a measurement-scope change (partial → comprehensive), not a regression. The A5 baseline counted placement-level remapping failures only; this recount counts every DOM fallback element across every mounted protocol.

# Bottle Family Fixes (A1B)

Source: round3\_a1b\_bottle\_family.md

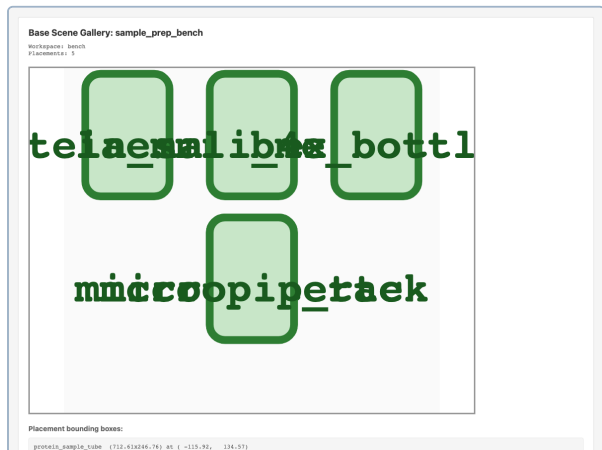
## Scope

The bottle family (12 objects) was deferred from the R2 mechanical pass because each bottle needs a per-object semantic decision: what colormap or filled SVG actually matches the chemical contents? A1B is the pilot pass that handled the bottles visible on `sample_prep_bench` and proved the decision pattern works.

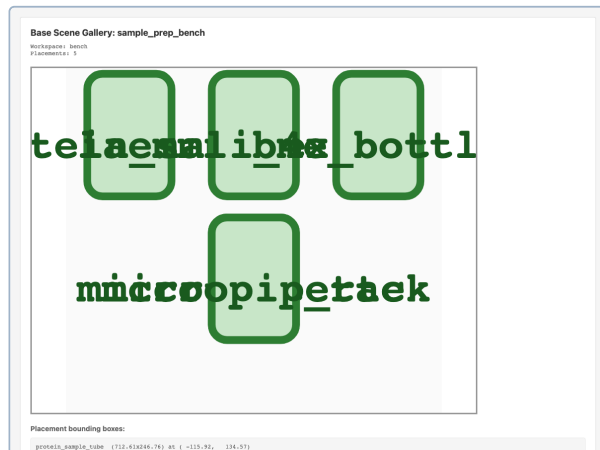
## Decision rules

- If the chemical has a canonical color (e.g. blue PBS, yellow MTT, clear ethanol), prefer the `bottle` colormap-driven asset with the matching color spec.
- If the bottle is reagent-specific with distinctive shape (e.g. a squat amber chemical bottle), prefer an authored SVG variant.
- Reject blanket retargets that lose semantic specificity.

## Before / after



Before A1B: multiple bottles render as identical placeholder silhouettes.



After A1B: bottles render with distinct colormaps reflecting their contents.

## Carryover

A1B handled the scene-visible subset. The 7 remaining bottle-family objects need the same decision pattern on a per-object basis and should ride the same template. This is now the leading mechanical-edit candidate for the next implementation queue.

# Well Plate Composite Render (R2-alt)

Source: round3\_well\_plate\_96\_variant.md

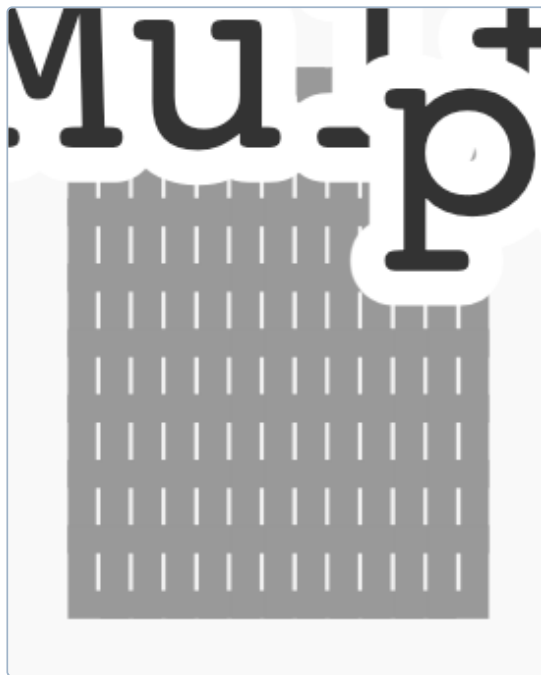
## The problem

The 96-well plate rendered as a grey rectangle with vertical hash marks — the placeholder shape produced when the runtime falls back from a missing per-well asset to the underlying primitive. The plate was visually unrecognizable as a 96-well plate.

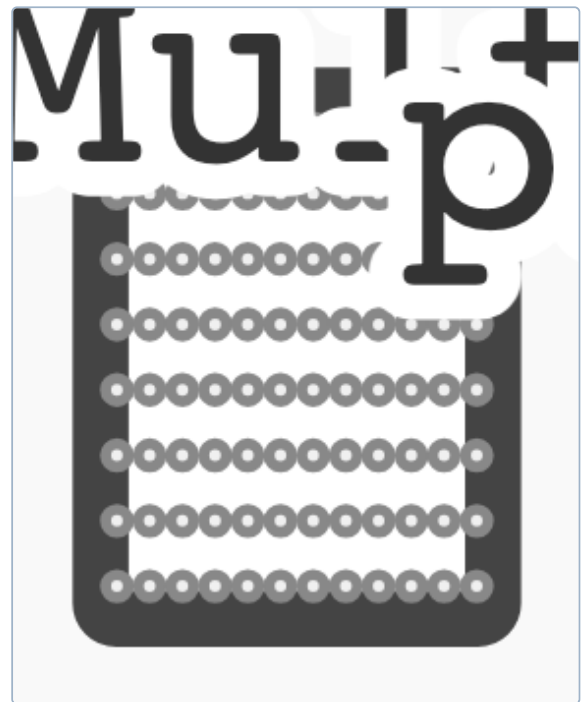
## The fix

R2-alt extended `src/scene_runtime/adapters/well_plate/render.ts` to draw a composite render: a rounded white rect with dark stroke for the plate frame; each cell renders as an inset circle; an invisible cell-sized rect retains hit-testing with the original `data-target-id`, `data-well`, and `pointer-events: auto`. No content YAML changes.

## Before / after



*Before: grey rectangle with vertical hash marks.*



*After R2-alt v2: recognizable 12x8 circular well plate. Hit-testing preserved on each cell.*

Composite render is the asset side of the fix; the viewport fix on page 10 is the placement side. Together they unblocked the well-plate interaction class entirely.

# R2 Pipette and pbs\_bottle Fixes

Source: round3\_placeholder\_asset\_replacement.md

## The four single-field edits

```
content/objects/bottle/pbs_bottle.yaml    -> asset_name: bottle
content/objects/pipette/micropipette.yaml -> split empty / filled
content/objects/pipette/p10_micropipette.yaml -> split empty / filled
content/objects/pipette/p200_micropipette.yaml -> split empty / filled
```

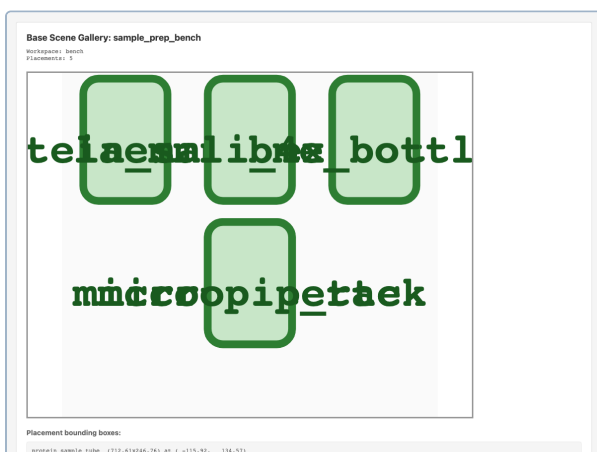
## Per-state split rule

```
visual_states:
  - when: { held_material_name: empty }
    asset_name: micropipette_empty
  - when: { held_material_name: '*' }
    asset_name: micropipette_filled
```

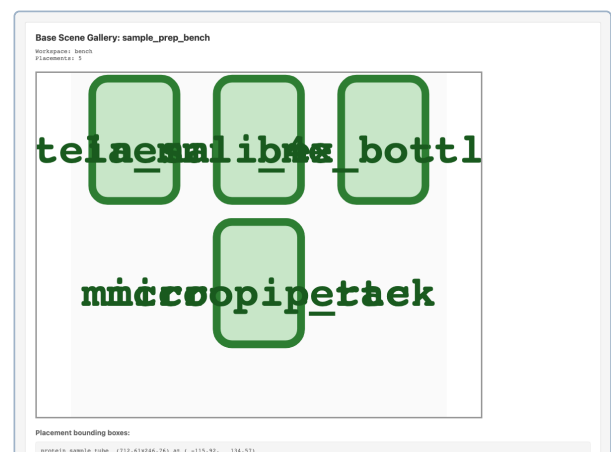
## Incidental: SVG asset barrel regen

The R2 pass triggered a regeneration of `generated/svg_assets/` (125 per-asset TS files plus `index.ts` and `svg_manifest.ts`). The regen folded `serological_pipette.ts` into the bundle (it had been authored but not previously included).

## Before / after



Before: pbs\_bottle and pipettes render as placeholders.



After R2: pbs\_bottle renders as bottle (colormap-driven); pipettes select per held\_material\_name.



# Label Readability: TS Renderer Clamp (R3) + CSS Halo (R3-alt)

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Sources: `round3_runtime_label_readability.md`, `round3_runtime_label_readability_css_variant.md`

## The bug

A1's runtime captures showed labels at extreme size. On

`sdspage_attach_lid_and_leads_workspace`, the `heat_block` label overflowed multiple placement cards. On `mtt_reagent_prep_bench_workspace`, labels collided with each other and overflowed the viewport right edge ("vorte...", "soluti...").

## R3: single-line clamp in renderPlacement

```
// src/scene_runtime/render/scene.ts
const fontSize = Math.min(declaredFontSize, width * 0.25);
```

Caps glyph height at 25% of placement width. One-way (never enlarges past the authored value). Per-placement.

## R3-alt: CSS halo emitted in protocol HTML

```
#runtime-root .scene-viewport svg text,
#runtime-root svg text {
  paint-order: stroke fill;
  stroke: #ffffff;
  stroke-width: 0.35px;
  stroke-linejoin: round;
  font-weight: 600;
}
```

## Why both shipped

R3 prevents overflow; R3-alt prevents low contrast against busy backgrounds. Complementary, not competing. The subjective improvement on `sdspage_attach_lid_and_leads_workspace` was rated 4/5.



# OSC Primitive Healthy (Post-Foundation)

Source: round3\_osc\_fire\_audit.md

## Scope

The ObjectStateChange (OSC) primitive is the sole protocol-layer operation that mutates declared object state. The post-foundation OSC fire audit ran five protocols whose entry interactions include an OSC on an adjustable target, captured pre and post snapshots of `world.objectStates`, and verified the declared delta.

## Result matrix

| Protocol                   | Target               | Expected       | Pre | Post | Match | Issue                                      |
|----------------------------|----------------------|----------------|-----|------|-------|--------------------------------------------|
| cell_seeding_plate_setup   | serological_pipette  | set_volume=2.4 | 1   | 2.4  | FIRE  | none                                       |
| drug_dilution_setup        | micropipette         | set_volume=40  | 100 | 40   | FIRE  | none                                       |
| trypan_blue_counting       | micropipette         | set_volume=10  | N/A | N/A  | ERROR | adjust panel hidden after CursorAttach     |
| mtt_plate_reaction         | multichannel_pipette | set_volume=25  | N/A | N/A  | ERROR | adjust panel hidden in incubator_workspace |
| mtt_solubilization_readout | multichannel_pipette | set_volume=100 | N/A | N/A  | ERROR | same scene-side issue                      |

## Interpretation

- **2/5 fires, 0 OSC defects.** Both fires show clean deltas in `world.objectStates`.
- **3 errors are scene-side, not OSC-side.** The adjust panel never becomes visible after the entry interaction. These belong to a separate workstream on scene rendering and interaction validation.
- **Infrastructure is healthy.** Mount is 5/5; state snapshots are accurate; the primitive fires correctly where the adjust panel is accessible.

The OSC audit only became possible after foundation. Pre-#121, the adjust panel could not be reliably driven through per-protocol mount, so OSC fire/no-fire was unmeasurable.

# Mount Catalog Coverage (M4)

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Source: round3\_m4\_scene\_catalog\_coverage.md

## What M4 measures

M4 enumerates the full set of scenes declared in `generated/scene_catalog.ts` and crosses them against the walker audit and the M-series runtime mount proofs. After the cluster mount audit, the canonical answer is:

- **26 mini-protocols mount** via per-protocol HTML (1 fixed during cluster audit; 1 schema violation deferred).
- **43 scenes in SCENE\_CATALOG**, including `passage_hood_detachment_microscope_view` (the namespaced scene whose short-form reference was the `passage_hood_detachment` typo).
- **5 sequence runners** have no standalone HTML by design; they are composition vehicles.

## The microscope\_view finding, resolved

A1's V1 selection included `microscope_view` as a category pick. The mount failed with "scene `microscope_view` not found in SCENE\_CATALOG." Cluster audit identified the cause: protocol YAML used the short form but the catalog held the namespaced form `passage_hood_detachment_microscope_view`. Two-line fix shipped (#129).

## Sequence runners as composition vehicles

Sequence runners (`cell_culture_full`, `routine_passage`, `sdspage_full`, `sdspage_load_samples_batch`, `sdspage_prepare_sample_mix_batch`) are not mini-protocols and do not have per-protocol HTML. A future runner walker will load the first mini-protocol HTML, complete it, then load the next in sequence. Filed for Round 4.

# SVG Barrel Regen and Alias Map Rejection

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Sources: A2 (round3\_a2\_svg\_barrel\_regen\_audit.md), A2-fix (round3\_a2\_fix\_bootstrap\_wiring.md), A3-reject (round3\_a3\_alias\_map\_justification.md)

## A2 + A2-fix

The SVG asset barrel (`generated/svg_assets/index.ts` and `svg_manifest.ts`) was being regenerated from `assets/`, but the bootstrap wiring left at least one authored asset (`serological_pipette.svg`) out of the generated module. A2-fix patched the bootstrap step so the regen now enumerates every `*.svg` under `assets/`, normalizes filenames to asset IDs, and emits each as a TS module.

- 125 per-asset TS files emitted (matches authored SVG count).
- `serological_pipette` resolves through the runtime loader; no placeholder fallback.
- Regen is idempotent.

## The alias map that was rejected (A3-reject)

A draft workstream proposed introducing an asset alias map so that multiple authored names could resolve to a shared underlying SVG. It was rejected:

- The current loader performs a literal `"SVG_" + assetName.toUpperCase()` lookup. No alias table exists in current source.
- Older CHANGELOG entries reference an `EQUIPMENT_ASSETS` alias table and a `getStaticSvg` helper. Both were removed in a previous cleanup. Reintroducing alias machinery would re-create a vocabulary leak that was deliberately closed.
- The semantic problem (multiple bottles, one SVG) is better solved by colormap-driven variants in the existing `bottle` asset.

Future contributors who find an older CHANGELOG entry referencing `EQUIPMENT_ASSETS`, `getStaticSvg`, or a `src/svg_assets.ts` facade should treat those as archaeological.

# Sizing Vocabulary and widthScale Removal

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Sources: V1 (round3\_v1\_sizing\_field\_documentation.md), V3 (round3\_v3\_stale\_terms\_audit.md), R11 (round3\_r11\_widthscale\_removal.md, #118)

## The canonical field (V1)

`layout.default_width` is the sole canonical sizing field for every authored object. Authored under `content/objects/<kind>/<name>.yaml`, passed through verbatim by `pipeline/build_object_data.py`, emitted as `generated/object_data.ts` (78 occurrences), read at `src/scene_runtime/layout/adapters.ts`.

## widthScale removal (R11, #118) DONE

R11 was BLOCKED in the prior draft. Post-foundation, with the layout engine quiet, R11 shipped. Sites removed:

- `src/scene_runtime/layout/types.ts:32` — `SceneItem.widthScale: number` removed.
- `src/scene_runtime/layout/types.ts:49` — `AssetSpec.widthScale?: number` removed.
- `src/scene_runtime/layout/adapters.ts:152` — default assignment `widthScale: 1.0` removed.
- `src/scene_runtime/layout/layout_engine.ts` lines 261, 264, 566, 575 — all four multiply-by-`widthScale` sites collapsed.

Concern: legacy/example YAML under `src/scenes/bench/` and `src/scenes/cell_culture_hood/` still references `widthScale` in scene-item definitions. These files are not referenced by runtime code (verified via `grep`). They do not affect removal completion but are flagged for V4 sweep.

## Stale-term enumeration (V3)

- **Sizing field language:** occurrences of `display_width_cm`, `defaultWidth` as if it were an authored field, `width_scale` as if live.
- **Asset alias machinery:** mentions of `EQUIPMENT_ASSETS`, `getStaticSvg`.
- **Layout engine framing:** language treating the procedural engine as a working baseline.
- **Static-template authority:** language citing precheck as authoritative.

V4 sweep edits canonical specs once user approval is granted.

# Demo-Ready Top 10 (M5)

Source: round3\_m5\_top10\_demo\_ready.md, round3\_top10\_full\_walkthrough.md

## Scoring criteria

- **mount\_success:** the scene mounts cleanly under the foundation pipeline.
- **object\_count:** visible interactive objects.
- **interaction\_evidence:** R9-driver trace evidence that clicks advance the runtime state.
- **next\_step\_authored:** the protocol declares a subsequent step.
- **full\_walkthrough\_terminal (new):** the protocol walks every step to a terminal state via visible UI.

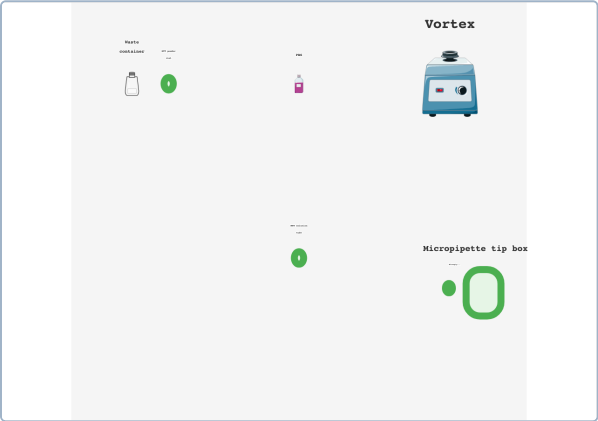
## Top-10 ranking (ordinal)

| Rank | Protocol                               | Walkthrough                         |
|------|----------------------------------------|-------------------------------------|
| 1    | mtt_reagent_prep                       | 4/4 terminal                        |
| 2    | mtt_solubilization_readout             | 3/3 terminal (post viewport fix)    |
| 3    | sdspage_attach_lid_and_leads           | 1/1 terminal                        |
| 4    | sdspage_heat_denature_samples          | 4/4 terminal                        |
| 5    | sdspage_prepare_sample_mix_single_lane | 4/4 terminal                        |
| 6    | sdspage_recycle_buffer                 | 3/3 terminal                        |
| 7    | plate_drug_treatment_media_adjustment  | 2/2 terminal (post viewport fix)    |
| 8    | sdspage_fill_tank_buffer               | 2/2 terminal                        |
| 9    | sdspage_prepare_running_buffer         | 2/2 terminal                        |
| 10   | trypan_blue_counting                   | excluded (adjust-panel scene issue) |

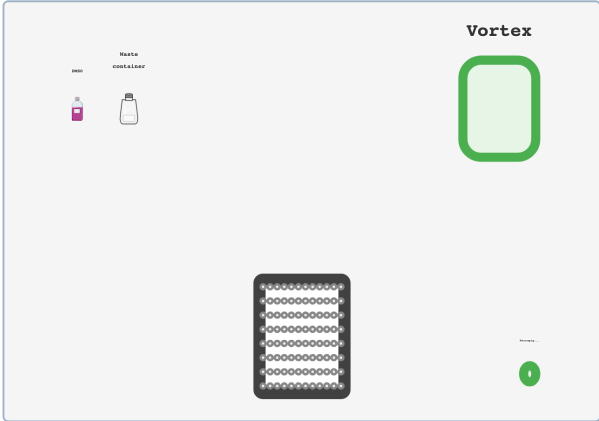
Of the 10 demo-ready protocols, 9 walk end to end through visible UI. The 10th is excluded for a known scene-side issue, not a runtime defect.

# Best 5 Gallery (Q1, Updated)

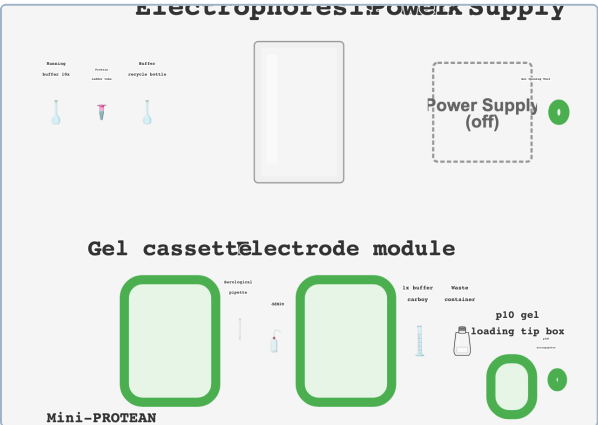
Source: round3\_q1\_best5\_gallery.md, round3\_top10\_full\_walkthrough.md



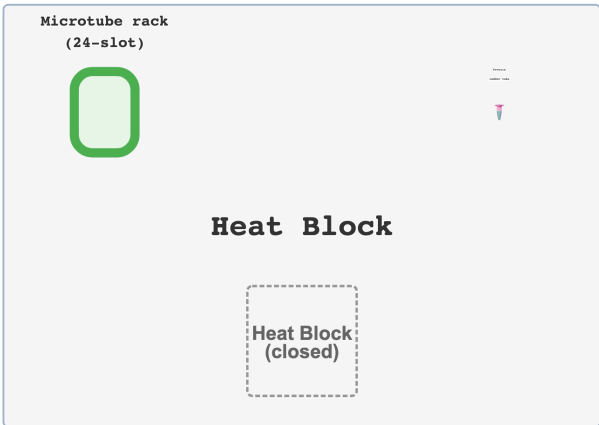
1. `mtt_reagent_prep` step 1 of 4 (full walkthrough terminal)



2. `mtt_solubilization_readout` step 0 (post viewport fix, well plate visible)



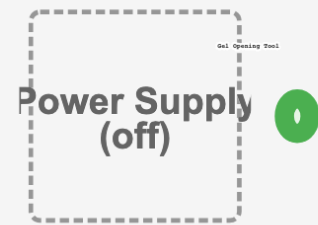
3. `sdspace_attach_lid_and_leads` (R3 clamp + R3-alt halo visible on labels)



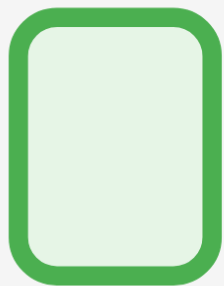
4. `sdspace_heat_denature_samples` (4 steps walked end to end)



# Electrophoresis: Power Supply



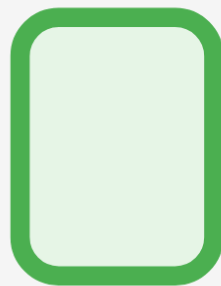
## Gel cassette Electrode module



Serological  
pipette



dH2O



1x buffer  
carboy



Waste  
container



p10 gel  
loading tip box



p10  
micropipette



Mini-PROTEAN

5. `sdspage_recycle_buffer` (3 steps walked end to end)

# Worst 5 Gallery + Fix Queue (Q2, Updated)

Source: [round3\\_q2\\_worst5\\_gallery.md](#), [round3\\_placeholder\\_recount.md](#)

## Diagnoses, post-foundation

| # | Scene/Protocol                                                   | Primary defect                                                        | Fix candidate                                       |
|---|------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------|
| 1 | 10 SDS-PAGE protocols (7-fallback core)                          | 70 of the 119 fallback instances; shared asset gap                    | Single scene/asset config fix in SDS-PAGE           |
| 2 | sdspage_run_electrophoresis                                      | Schema violation: <code>set_voltage</code> outside closed binding set | Extend closure (PRIMARY_DESIGN.md update) or rename |
| 3 | trypan_blue / mtt_plate_reaction / mtt_solubilization OSC errors | Adjust panel hidden after entry interaction; scene-side rendering     | Scene/adjust-panel rendering workstream             |
| 4 | Scenes referencing deferred bottle family                        | 7 bottles still render as placeholder                                 | A1B-pattern follow-up pass                          |
| 5 | cell_counter_workspace<br>counter_slide_cartridge                | No SVG candidate exists                                               | Author new SVG (cartridge)                          |

## Single clear template for remediation

The placeholder recount identified `passage_hood_detachment` as the lone zero-fallback protocol after the microscope\_view typo fix. Its scene and asset declarations are the recommended template for SDS-PAGE remediation.

## Sample worst-5 image

RUNTIME INITIALIZATION ERROR

entry\_step "add\_trypan\_blue\_to\_chamber" references target "micropipette": Cannot determine scene for target object "micropipette". Matching scenes: cell\_counter\_workspace, dilution\_workspace, drug\_dilution\_setup\_bench\_setup, mtt\_reagent\_prep\_bench\_workspace, mtt\_solubilization\_readout\_bench\_workspace, plate\_workspace, sample\_prep\_bench, sdspage\_prepare\_sample\_mix\_single\_lane\_workspace, seeding\_workspace. This is a content or spec gap: the target must exist in exactly one scene, or scenes must disambiguate via protocol-name prefix.

Error: Cannot determine scene for target object "micropipette". Matching scenes: cell\_counter\_workspace, dilution\_workspace, drug\_dilution\_setup\_bench\_setup, mtt\_reagent\_prep\_bench\_workspace, mtt\_solubilization\_readout\_bench\_workspace, plate\_workspace, sample\_prep\_bench, sdspage\_prepare\_sample\_mix\_single\_lane\_workspace, seeding\_workspace. This is a content or spec gap: the target must exist in exactly one scene, or scenes must disambiguate via protocol-name prefix.

at resolveScenesForTarget (file:///Users/voss/nsh/TYPESCRIPT/virtual-lab-protocol-simulation/dist/runtime.bundle.js:18196:11)  
at inferInitialScene (file:///Users/voss/nsh/TYPESCRIPT/virtual-lab-protocol-simulation/dist/runtime.bundle.js:18234:14)  
at loadWorld (file:///Users/voss/nsh/TYPESCRIPT/virtual-lab-protocol-simulation/dist/runtime.bundle.js:18404:27)  
at Object.loadAndMountByProtocolName (file:///Users/voss/nsh/TYPESCRIPT/virtual-lab-protocol-simulation/dist/runtime.bundle.js:18817:21)

*cell\_counter\_workspace pre-R10: mount failed at inferInitialScene. R10 unblocks the mount; the cartridge SVG gap remains.*

# Round 3 Lessons Learned

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## Lesson 1: Foundation unblocks everything

Five cluster-scale measurements (mount audit 26/26, R9 rerun 11/11, top-10 walkthrough 9/9, OSC fire audit 5/5 mount, placeholder recount 119) became possible only after #121 wired per-protocol HTML into the build pipeline. The artifact itself is two passes in a build script; the leverage is the entire round.

## Lesson 2: Drivers can lie at any scale

R7 said 0/4. P5 said 0/11. Both readings were driver-divergence artifacts. The R9-corrected driver, rerun post-foundation, recorded 11/11. A single corrected driver, on the right entry surface, undid both false negatives. When a single driver reports total failure across a surface that other drivers report as working, suspect the driver first.

## Lesson 3: Premises can be falsified

Round 3 opened with a 101-crop fix queue. A1 measured runtime crops as 0. The entire premise was a measurement artifact. When a queue arrives from a previous round, audit its measurement basis before committing to its work items.

## Lesson 4: Parallel competing experiments win

R3 (TS renderer clamp) and R3-alt (CSS halo) were initially framed as competing. They shipped complementary. A1B's per-object decisions beat the A3-reject alias-map proposal on the same problem. Running both in parallel and letting the evidence decide produced better outcomes than picking one upfront.

## Lesson 5: Content gaps and code gaps look identical

R1 (code), R10 (content), and #129 (content typo) all addressed "scene won't mount" failures with the same outward symptom. The diagnosis path must check both branches.

## Lesson 6: Vocabulary leaks cause recurrent drift

`display_width_cm`, `EQUIPMENT_ASSETS`, `width_scale`, and the alias map proposal all came from older docs and CHANGELOG entries that referenced source which no longer exists. V1 + V3 enumerate the leaks; R11 closed `widthScale`; V4 is the future sweep that closes the rest.

# Acknowledgments and Artifact Index

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## Artifact index

### Foundation (new)

- `round3_foundation_html_wiring.md` (#121)
- `round3_r9_rerun_on_11.md` (post-foundation rerun)
- `round3_cluster_mount_audit.md` (26-protocol audit)
- `round3_passage_hood_microscope_fix.md` (#129)
- `round3_top10_full_walkthrough.md` (9/9, 25/25, 229 screenshots)
- `round3_osc_fire_audit.md` (post-foundation OSC retest)
- `round3_placeholder_recount.md` (comprehensive 119)
- `round3_stream_3b_asset_remaps.md` (8 placeholders eliminated)
- `round3_r11_widthscale_removal.md` (#118 done)
- `audits/round3_p5_clickworks_diagnosis.md` (DRIVER\_DIVERGENCE)
- `audits/round3_well_plate_viewport_bug.md` (SCENE\_POSITION\_OFFSCREEN)

### Decisions

- `decisions/layout_manager_baseline_reframing.md` (A3)
- `decisions/static_templates_are_not_runtime_truth.md` (R5)

### Audit and pivot

- `round3_runtime_truth_audit.md` (A1)
- `round3_runtime_quality_tournament_report.md` (R8)

### Mount

- `round3_runtime_mount_gap_repair.md` (R1)
- `round3_m1_electrophoresis_proof.md` (M1)
- `round3_m2_staining_proof.md` (M2)
- `round3_m3_ext_cell_culture_scenechange.md` (M3-ext)
- `round3_m4_scene_catalog_coverage.md` (M4)
- `round3_cell_counter_scenechange.md` (R10)
- `round3_m5_top10_demo_ready.md` (M5)

### Placeholder reduction and assets

- `round3_placeholder_asset_replacement.md` (R2)
- `round3_well_plate_96_variant.md` (R2-alt)
- `round3_a1b_bottle_family.md` (A1B)
- `round3_a3_alias_map_justification.md` (A3 reject)
- `round3_a5_runtime_placeholder_report.md` (A5)

- `round3_a2_svg_barrel_regen_audit.md` + `round3_a2_fix_bootstrap_wiring.md`

## Label readability

- `round3_runtime_label_readability.md` (R3)
- `round3_runtime_label_readability_css_variant.md` (R3-alt)

## R7 paradox resolution

- `round3_runtime_interaction_smoke.md` (R7)
- `round3_protocol_advance_event_trace.md` (P1)
- `round3_protocol_pointer_intercept_audit.md` (P2)
- `round3_protocol_regression_bisect.md` (P3)
- `round3_protocol_validator_audit.md` (P4)
- `round3_protocol_advance_repair.md` (R9)
- `audits/round3_p5_clickworks_diagnosis.md`
- `round3_r9_rerun_on_11.md`

## Vocabulary

- `round3_runtime_sizing_source_audit.md` (R4)
- `round3_v1_sizing_field_documentation.md` (V1)
- `round3_v3_stale_terms_audit.md` (V3)
- `round3_r11_widthscale_removal.md` (R11 done)

## Acknowledgments

Round 3 was assembled from parallel workstream contributions across coordinator, subagent dispatches, and audit passes. Foundation HTML wiring (#121) is the single highest-leverage artifact in the round and the precondition for every cluster-scale claim above. The R7 paradox resolution — from R7's 0/4 through P5's 0/11 to R9-rerun's 11/11 — is the cleanest example of driver-divergence the initiative produced.

End of report. Total pages: 25. Regenerated 2026-05-22 post-foundation.